Process Safety Essentials

A practical quick-reference guide to loss of containment and barriers

01 / OVERVIEW

01 Identify the hazard

Hazard: a source with the potential for harm, such as a pressurised hydrocarbon system. Threat: a condition that may lead to loss of control, such as inadequate gasket compression or an incorrect operation.

02 Name the top event

The top event is loss of control. Example: hydrocarbon escapes from a pressurised flange. Make the statement specific and observable.

03 Think about consequences

After loss of containment, possible consequences include fire, explosion, exposure, pollution and production interruption. The actual outcome depends on release conditions and safeguards.

04 Challenge assumptions

Strong personal safety performance does not guarantee process-safety performance. Monitor hazardous conditions, barrier availability and critical deviations.

Barriers & verification

Use before start-up, after maintenance and during abnormal conditions

02 / FIELD CHECKLIST

05 Preventive barriers

Before the top event: correct joint assembly, controlled bolt tightening, suitable materials, inspection, verified isolation, safe operating limits and pre-service tests.

06 Mitigative barriers

After the top event: prompt detection, alarms, emergency isolation, competent response, fire protection, evacuation and spill control.

07 Verify critical controls

Who owns the barrier? What evidence shows it is available and effective? When was it tested? What escalation is required if it fails or is unavailable?

08 Respond to abnormal conditions

Recognise and communicate alarms or deviations, follow site-specific operating procedures, and use the authorised escalation process. Never bypass a critical safeguard informally.

DISCUSSION PROMPT

Where could a single weak control cause loss of containment on your site?